```
result = session.run(slice, feed_dict={image: raw_image_
data})
print(result.shape)

plt.imshow(result)
plt.show()
```

Variables

In TensorFlow, variables are used to manage data. A TensorFlow variable is the best way to represent shared and persistent tensor states to be manipulated by a program. Specific operations are required to read and modify the values of this tensor. Since a *tf.Variable* exists outside the context of a single *session.run* call, modified values are visible across multiple *tf.Session*, and multiple users can view the same values. These variables can be accessed via the *tf.Variable* class objects.

Many parts of the TensorFlow library use this facility. For example, when a variable is created, it is added by default to collections representing global variables and trainable variables. When in later stages *tf.train.Saver* or *tf.train.Optimizer* are created, the variables in these collections are used as the default arguments.

Variable initialisers

Global tensor variables can be initialised using the following TensorFlow methods:

```
    tf.global_variables_initializer
tf.initializers.global_variables
tf.initializers.global_variables()
```

There is an alternative shortcut to implicitly initialise a variable list. The shortcut is *tf.variables_initializer(var_list, name='init')*. If there is no definition of the variables before calling *tf.global_variables_initializer*, the variable list is essentially empty.

The code below illustrates this:

```
import tensorflow as tf

with tf.Graph().as_default():
    # Nothing is printed
    for v in tf.global_variables():
    print("List Global variable 1", v,"\n")


init_op = tf.global_variables_initializer()

  a = tf.Variable(0)
  b = tf.Variable(0)
  c = tf.Variable(0)

init_op = tf.global_variables_initializer()
 # 3 Variables are printed here
for v in tf.global_variables():
     print("List Global Variable 2", v,"\n")

with tf.Session() as sess:
     sess.run(init_op)
     print( "List session",sess.run(a),"\n")
```

Execution of this script will not generate the variable list of the first loop, whereas the second for-loop will iterate four times over the global variable list and will display the variable graph in four lines.

```
init_op = tf.global_variables_initializer()
```

After this, the following lines:

```
  a = tf.Variable(0)
  b = tf.Variable(0)
  c = tf.Variable(0)
  d = tf.Variable(0)
```

…will initialise all the global variables lists for further processing of the graph.

A simple mathematical operation

Here is an example of a simple TensorFlow mathematical operation using variables a and b:

```
a = tf.Variable([4])
b = tf.Variable([7])
with tf.Session() as session:
     session.run(tf.global_variables_initializer())
     b = a + b
     result = session.run(a)
     print(a)
     result = session.run(b)
     print(b)
     print(session.run(a))
      print(session.run(b))
```

TensorFlow software is a new programming concept designed on graph based computing. This article is an introductory tutorial to understand this open source programming environment. The building blocks of this computing environment have been discussed with simple examples. The reader will hopefully get an idea of TensorFlow programming from this brief introduction. END

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